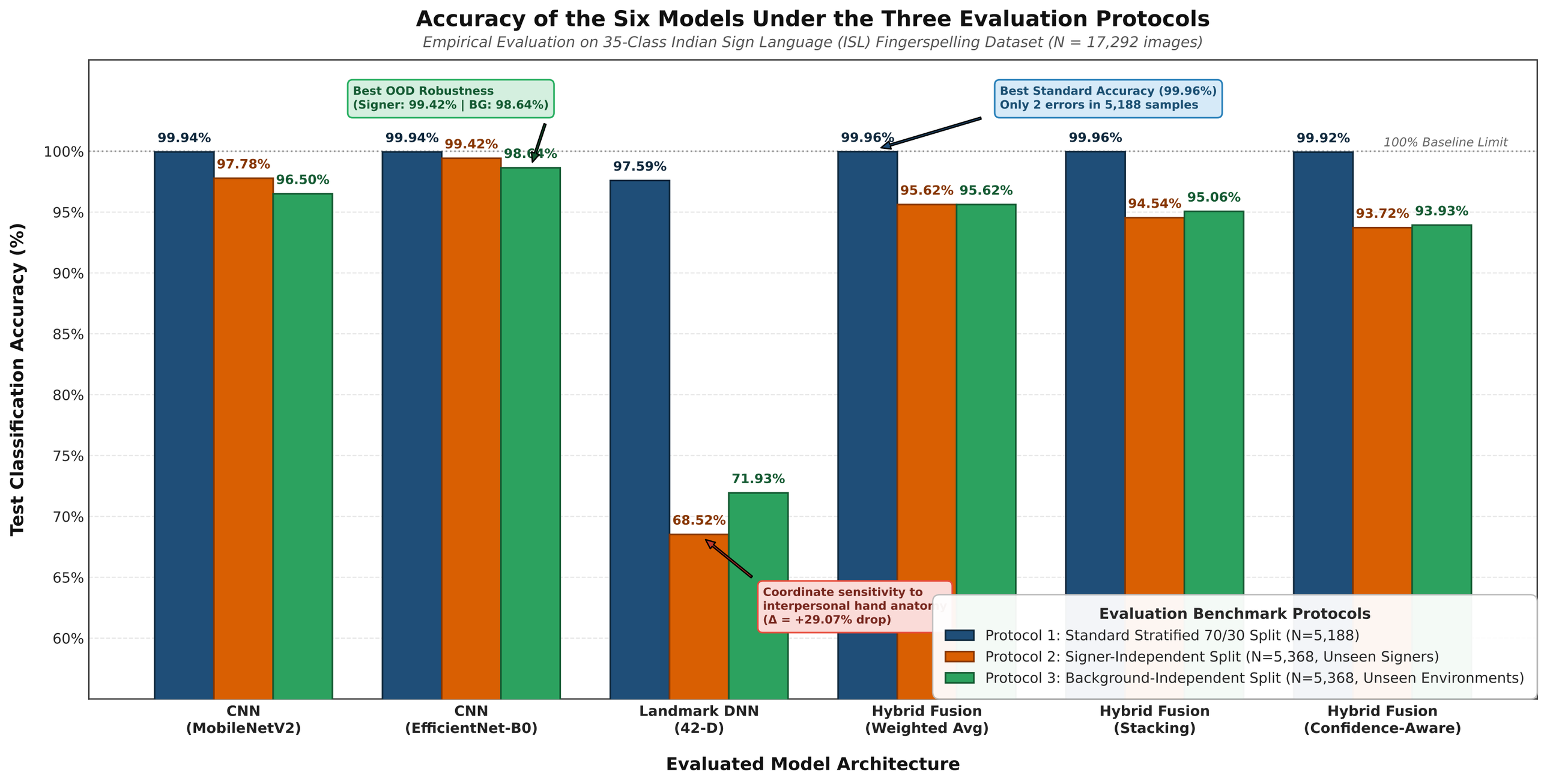

Accuracy of the Six Models Under the Three Evaluation Protocols
Empirical Evaluation on 35-Class Indian Sign Language (ISL) Fingerspelling Dataset (N = 17,292 images)
Test Classification Accuracy (%)
100%
95%
90%
85%
80%
75%
70%
65%
60%
100% Baseline Limit
Best OOD Robustness
(Signer: 99.42% | BG: 98.64%)
Best Standard Accuracy (99.96%)
Only 2 errors in 5,188 samples
99.94%
97.78%
96.50%
99.94%
99.42%
98.64%
97.59%
68.52%
71.93%
99.96%
95.62%
95.62%
99.96%
94.54%
95.06%
99.92%
93.72%
93.93%
Coordinate sensitivity to
interpersonal hand anatomy
(Δ = +29.07% drop)
Evaluation Benchmark Protocols
Protocol 1: Standard Stratified 70/30 Split (N=5,188)
Protocol 2: Signer-Independent Split (N=5,368, Unseen Signers)
Protocol 3: Background-Independent Split (N=5,368, Unseen Environments)
CNN
(MobileNetV2)
CNN
(EfficientNet-B0)
Landmark DNN
(42-D)
Hybrid Fusion
(Weighted Avg)
Hybrid Fusion
(Stacking)
Hybrid Fusion
(Confidence-Aware)
Evaluated Model Architecture